LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

# Lesson 8: Models and Agents

Module 3 | 45 minutes | Modelling | Grades 6–8 · ages 11–14

**Objective:** Build a numeric model that changes over time, and an agent-based model where many clones follow simple rules.

**Success criteria**

- I can build a step-by-step model.
- I can build an agent-based model with clones.
- I can explain what a model leaves out.

## Key vocabulary

- **model**: A simplified version of reality used to predict.
- **time step**: One update of the model (e.g. a year).
- **agent**: An individual following rules (often a clone).
- **emergence**: Patterns that arise from many simple rules.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

## Timed lesson plan (45 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Human agents (5 min): learners walk with one rule (“keep 1 m from everyone”) and watch the pattern. |
| 5 min | **Watch** | Play the lesson video with captions; pause at each heading and ask a question. |
| 8 min | **Build along** | Rabbits and particles: learners follow the steps. |
| 14 min | **Perform** | Code Lab: population model, Code Lab: agent particles |
| 9 min | **Probe** | Debug the model, activity |
| 4 min | **Wrap-up** | Exit ticket: Name one thing your population model ignores. |

## Unplugged warm-up

Human agents (5 min): learners walk with one rule (“keep 1 m from everyone”) and watch the pattern.

## Teaching the idea

Discrete-time models update state per step (exponential growth with a carrying-capacity clamp). Agent-based models simulate individuals with local rules; macro patterns emerge. Models are simplifications — always question assumptions.

**Analogy:** A map: useful because it leaves things out.

**Live demo:** Run the model without and with the limit; plot pop each year on the board.

## Questions to ask

| Ask | Listen for |
|---|---|
| What does the model leave out? | Predators, disease, weather. |
| Why is the limit inside the loop? | It applies every step. |
| What emerges from the particles? | They spread evenly, like gas. |

## Differentiation

**Support**
Provide the growth loop; learners add the limit.

**Stretch**
Make particles change colour when they touch another particle.

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| A model is the same as reality | Models simplify and can be wrong. |
| Agents need complicated code | Simple rules make complex patterns. |

## Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

**Exit ticket**
Name one thing your population model ignores.

**Homework**
Find a news story that uses a model or simulation. What might it leave out?

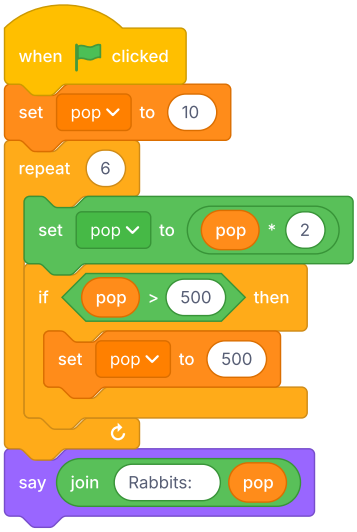
# Answer key — Kitabash activities

## 1. Code Lab: population model (perform · code lab)

**Goals checked:** Doubles each year · Respects the limit · Reports it

**One solution:**

**Scientist**

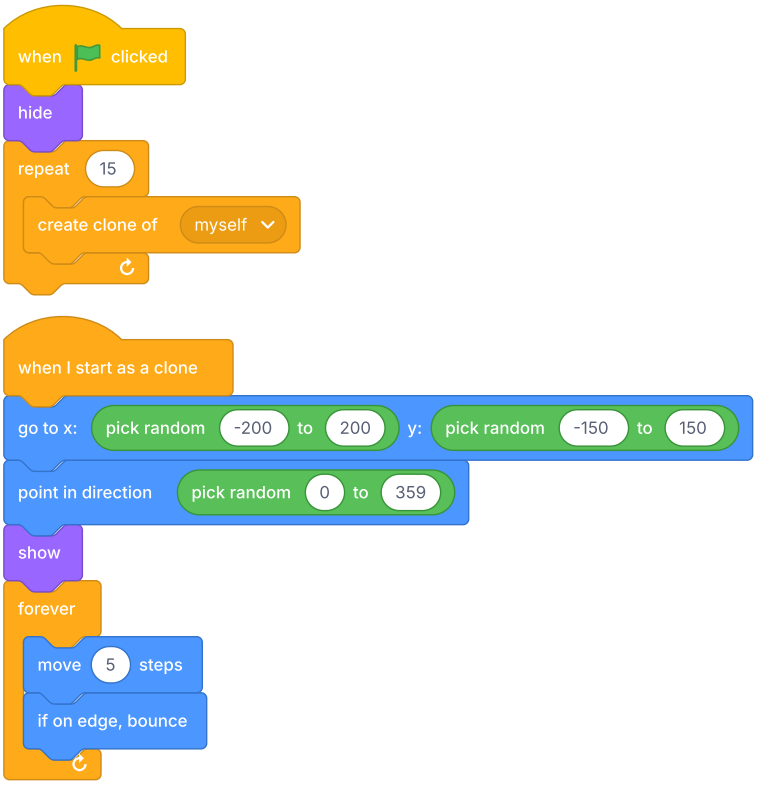


### 2. Code Lab: agent particles (perform · code lab)

**Goals checked:** 15 agents · Random start · Agents move and bounce

**One solution:**

**Particle**




### 3. Debug the model (probe · multiple choice)

**Answer:** The limit check is outside the loop, so it only runs once at the start

### 4. Explain your thinking (probe · free response)

**Look for:** model, rule, ignores. **Example:** The model ignores foxes and disease. I could add foxes as a rule: each year, set pop to pop minus foxes times 3, so more foxes means fewer rabbits. That would make the model more realistic.

## Answer key — worksheet

1. **Key words:** model = A simplified version of reality used to predict. · time step = One update of the model (e.g. a year). · agent = An individual following rules (often a clone). · emergence = Patterns that arise from many simple rules.
2. **Plan your code:** Goals: Doubles each year; Respects the limit; Reports it.
3. **Plan your code:** Goals: 15 agents; Random start; Agents move and bounce.
4. **Debug the model:** A — Rules that apply every year must be inside the yearly loop.
5. **Explain your thinking:** The model ignores foxes and disease. I could add foxes as a rule: each year, set pop to pop minus foxes times 3, so more foxes means fewer rabbits. That would make

the model more realistic.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.